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A General Measure of User Computing Satisfaction

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Abstract — *User satisfaction with computers is an increasingly important topic in view of the explosive growth in the number of organizational personnel who use computers in their work and the resulting need to evaluate the effectiveness of such usage. Doll and Torkzadeh (1988) presented an instrument to measure user satisfaction with specific computer applications. In the present study, data from 776 organizational knowledge workers supported the construct validity and reliability of Doll and Torkzadeh's instrument when used as a general measure of user satisfaction. This study also investigated the relationship of user satisfaction and computer attitudes, computer anxiety, computer skill, and computer usage to contribute to the nomological network of the user satisfaction construct.*

The explosive growth in the number of organizational personnel who directly use computers in their work has continued for the past decade (e.g., see Benjamin, 1982). Recent studies have shown that most organizational knowledge workers make direct use of computers in the course of their work. These people use computers for a large variety of different tasks or applications (e.g., word processing, spreadsheets, statistics, databases, etc., Rainer & Harrison, 1993;

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McLean & Kappelman, 1993). The large number of people using computers in their work means that there is a need to evaluate the effectiveness of such usage.

"User satisfaction is probably the most widely used single measure of information systems success" (Delone & McLean, 1992). Melone (1990) proposes a strategy for developing a theoretical base for understanding the user satisfaction construct. The strategy calls for focusing attention on user attitudes, which is a broader construct than user satisfaction.

The general measure of user satisfaction may be theoretically grounded in the attitude-behavior theory of Fishbein and Ajzen (Fishbein, 1979; Fishbein & Ajzen, 1975). Their theory of reasoned action postulates that beliefs about an object lead to an attitude about it. This attitude leads to behavioral intentions regarding the object. These intentions, in turn, affect the actual behaviors toward the object. Finally, there is a feedback loop in which behavioral experiences can modify the beliefs about the object.

The theory of reasoned action has important implications for computer use. The theory suggests that user satisfaction, an attitude, affects users' behavioral intentions toward computer use, which affect users' actual usage of computers. Because computer use is critically important in knowledge work, user satisfaction becomes an important construct which must be understood and measured. A general measure for user satisfaction is necessary to allow for diversity among users and diversity in the computer tasks they perform.

The purpose of this paper is to examine the End User Computing Satisfaction Instrument of Doll and Torkzadeh (1991) as a general measure of user satisfaction. We utilize this instrument in a field setting and assess its validity and reliability.

BACKGROUND

User satisfaction is one of the most widely researched topics in the information systems field (Table 1). Locke (1976) defined satisfaction as an emotional response or affect toward an object. Information systems researchers have used similar definitions of satisfaction. Bailey and Pearson (1983) defined satisfaction as the sum of feelings or attitudes, both positive and negative, toward the spectrum of factors affecting a specific situation. Doll and Torkzadeh (1988) also defined satisfaction as an affective attitude. Galetta and Lederer (1989) concluded that satisfaction includes both perceptions and attitudes.

Melone (1990) defined a user attitude as a "predisposition to respond favorably or unfavorably to a computer system, application, system staff member, or a process related to the use of the system or application" (p. 81). She proposed that end user satisfaction defined as an attitude can be theoretically justified in Fishbein and Ajzen's theory of reasoned action (Fishbein & Ajzen, 1975). The theory of reasoned action suggests that an individual's behavior can be predicted from the individual's attitude toward (a) the behavioral action, and (b) the social norms that influence the probability of performing the behavior. Melone (1990) concluded that the theory of reasoned action provides a theoretical basis for concluding that an individual's perceptions of the computer system and related activities are predictive of success of the computer system. The preceding discussion suggests that end user satisfaction is an affective

Table 1. Chronological Progression of User Satisfaction Studies

Study Author	Year
Swanson, E. B.	1974
Zmud, R. W.	1979
Larcker, D. F., & Lessig, V. P.	1980
Bailey, J. E., & Pearson, S. W.	1983
Ives, B., Olson, M., & Baroudi, J. J.	1983
Ives, B., & Olson, M. H.	1984
Davis, J. C.	1985
Sanders, G. L., & Courtney, J. F.	1985
Baroudi, J. J., Olson, M., & Ives, B.	1986
Goodhue, D.	1986
Rushinek, A., & Rushinek, S. F.	1986
Iivari, J.	1987
Raymond, L.	1987
Baroudi, J. J., & Orlikowski, W. J.	1988
Bergeron, F., & Berube, C.	1988
Doll, W. J. and Torkzadeh, G.	1988
Montazemi, A. R.	1988
Davis, F.	1989
Davis, F., Bagozzi, R., & Warshaw, P.	1989
Galetta, D. F., & Lederer, A. L.	1989
Igbaria, M., & Parasuraman, S.	1989
Benbasat, I.	1990
Igbaria, M., & Nachman, S. A.	1990
Joshi, K.	1990
Melone, N. P.	1990
Doll, W. J., & Torkzadeh, G.	1991
Etezadi-Amoli, J., & Farhoomand, A. F.	1991
Moore, G., & Benbasat, I.	1991
Torkzadeh, G., & Doll, W. J.	1991

attitude toward all the various activities surrounding an end user's interaction with a computer-based information system.

Rainer and Harrison (1993) defined end user computing at the individual level of analysis as the direct, individual use of computers encompassing all the computer-related activities required or necessary to accomplish one's job. As suggested by Carr (1988, p. 70), typical end user computing tasks include spreadsheet applications, database management, word processing, programming, data analysis, graphics, communications, data query, and memory assistance. These tasks are typically combined to achieve management information and decision support functions.

Basing the definition of end user satisfaction on the end user computing construct, end user satisfaction is an individual's attitude toward the use of computers spanning all job-related computing activities (Rainer & Harrison, 1993). Consequently, an empirical operationalization of the end user satisfaction construct should take all these activities into account.

Ajzen and Fishbein (1977) argued that attitudinal and behavioral entities consist of four elements: (a) action, (b) target at which the action is performed, (c) context in which the action is performed, and (d) time in which the action is performed. They concluded that examination of the target and action elements is sufficient for attitude-behavior correspondence. They noted "When the criterion is an index based on observations of heterogeneous behaviors with respect to a given target, only the target element is specified and the resulting measure may be called a multiple-act criterion" (p. 891). Consequently, the user satisfaction construct is a multiple-act criterion. That is, because users may employ many

different computer applications for diverse tasks in their work, satisfaction with a wide variety of heterogeneous behaviors should be acknowledged. Therefore, a general measure is necessary to examine all behaviors with respect to the target element (i.e., the computer system).

The End User Computing Satisfaction Instrument

Early research on user satisfaction developed measurement instruments that focused on general user satisfaction (e.g., Bailey & Pearson, 1983; Ives & Olson, 1984). However, several subsequent studies pointed out weaknesses in these instruments (e.g., Galetta & Lederer, 1989; Treacy, 1985).

In response to the perceived weaknesses in prior instruments, Doll and Torkzadeh (1988) developed the 12-item End User Computing Satisfaction Instrument (EUCSI). Their instrument was designed to measure user satisfaction with specific applications and was sent to "major users" (Doll & Torkzadeh, 1988, p. 264) in 44 organizations.

The use of the EUCSI as a general measure of user satisfaction has several rationales. First, Doll and Torkzadeh stated that "the items . . . were selected because they were closely related to each other (i.e., all items were thought to be measures of the same . . . satisfaction construct)" (Doll & Torkzadeh, 1988, p. 265). Second, 8 of Doll and Torkzadeh's 12 items use the term "system". A user could perceive the term "system" to nonspecifically encompass all the computer-based information systems and applications that he/she might encounter. Third, earlier measures of user satisfaction focused on general satisfaction and the items of the EUCSI (with the exception of ease of use) are found in these earlier instruments. Fourth, Doll and Torkzadeh (1988) surveyed "major users . . . (which) may have failed to identify a few major end users, especially microcomputer users" (p. 264). Because the majority of organizational personnel directly use computers (particularly microcomputers), a user satisfaction instrument should be distributed to all organizational personnel, from nonsophisticated computer users to sophisticated computer users. In this case, a problem arises in defining the term "application" for heterogeneous users, certainly for nonsophisticated computer users. Further, users typically use many different applications. Therefore, asking such diverse users to consider their computer systems in general may be a valid method of measuring satisfaction as the user perceives it, regardless of his/her computer skill, sophistication, or particular application.

This study employs the EUCSI as a general measure of user satisfaction. Accordingly, respondents were asked to answer each of the 12 statements by considering all the computer-based information systems and applications they were currently using in their jobs. In other words, rather than asking respondents to assess their satisfaction with a specific application as Doll and Torkzadeh (1988) specified, the study asked respondents to assess their satisfaction across all applications. The unit of analysis in this study was the individual rather than the application.

The purpose of this study is to provide evidence for the use of the EUCSI as a general measure of user satisfaction. Personnel throughout an entire organization were surveyed to gather data on their satisfaction with their computer systems in general. Respondents exhibited the complete spectrum of computer sophistication and skill, from novice to expert. The findings support the

instrument's construct validity and reliability when it is used as a general measure of user satisfaction.

METHODOLOGY

The authors mailed a questionnaire regarding various aspects of computer use to the 3,488 salaried personnel of a university. The first part of the questionnaire gathered demographic data on the respondents. The remainder of the questionnaire included a variety of measures, one of which was the EUCSI.

In order to use the EUCSI as a general measure of user satisfaction, it was important to meet two criteria. First, the sample should include a variety of users who interact with a large set of diverse applications. Second, proper instructions should be given to the respondents. The first criterion is addressed in the discussion of the sample below. The second criterion was met by asking respondents to answer each item by considering, as a whole, all the computer-based information systems and applications they were currently using in their jobs.

It is important to note that Doll and Torkzadeh (1988) surveyed persons who could be classified in this study as advanced end users, end user facilitators, and end user infrastructure personnel. Beginning and intermediate end users were omitted from their study, in spite of the fact that these people constitute the majority of organizational personnel who use computers in their work (see also, Rockart & Flannery, 1983). These users are nonsophisticated compared with those surveyed by Doll and Torkzadeh. It seems logical to ask these users to consider their computer systems and applications in general, because they may not know how to differentiate between applications or, indeed, what an application is.

The Sample

A total of 776 usable responses were received, giving a response rate of 22.3%. A follow-up mailing was prohibited by university regulations, therefore non-response bias was checked in the following manner. After the cutoff date, 41 questionnaires were received and these were held to check nonresponse bias (Babbie, 1973; Kerlinger, 1973; Nunnally, 1978). No significant differences were revealed by t tests which compared the demographic variables between the first and second groups of respondents. The conclusion was reached that nonresponse bias was not evident and that the results could be generalized to the university population of knowledge workers.

The sample included respondents from four university job categories: faculty (43%), technical (7%), administrative (20%), and clerical (30%). The t tests conducted indicated no significant differences between the percentages of each category in the population and percentages of each category in the sample. A total of 51% of the respondents were male and 72% possessed at least a bachelor's degree. The mean age of the respondents was 38 years, and they averaged 7.5 years of hands-on experience with computers.

Respondents from every department in the organization were included in the sample. They represented all ranks in each job category: clerical (Secretary I to Administrative Assistant IV); technical (Research Technician I to Research

Technician VIII); faculty (Instructor to Professor); and administrative (Department Head to Vice President).

The respondents used a wide variety of hardware and software. The hardware included all types of computers, from supercomputers to microcomputers. Respondents reported that they utilized over 150 different application software packages, including word processing, spreadsheet, graphics, database management, communication, and statistics. They employed these packages in monitor, exception, inquiry, and analysis applications.

DATA ANALYSIS AND FINDINGS

Construct validity is established by showing that the measure is an appropriate operational definition of the construct it purports to be measuring (Cronbach & Meehl, 1955; Stone, 1978). Factor analysis, reliability analysis, and known groups analysis are three methods of construct validation (Campbell, 1976; Kim & Mueller, 1982).

Factor Analysis

Exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) are two major approaches to factor analysis. In areas where little research has been done, EFA can suggest underlying patterns in the data (Gerbing & Anderson, 1988). If, however, hypotheses about plausible model structures exist, then CFA may be used to test these models (Bollen, 1989). That is, CFA requires a specified, initial model.

Doll and Torkzadeh (1988, p. 268) used EFA with an orthogonal rotation to force a five-factor solution to derive their model of user satisfaction. They labeled these five factors content, accuracy, ease of use, timeliness, and format. The content construct has four indicator variables, while the remaining four constructs have two indicators each.

Doll and Torkzadeh's model may therefore be used as the initially specified model CFA with an important difference. Their EFA, with its orthogonal rotation, produced five underlying constructs that are assumed to be uncorrelated. This finding is open to question, particularly when the meanings of the constructs are examined. It is important to note that the CFA model used in this study postulates that the five constructs are, in fact, correlated.

Figure 1 displays the hypothesized model, which is a second-order confirmatory factor analysis (Bollen, 1989). This study uses LISREL 7 (Joreskog & Sorbom, 1988) to perform the CFA which tests the goodness-of-fit of the hypothesized model to our data (Bollen, 1989).

A variety of statistics may be used to test goodness-of-fit of a model to the data. These indices may be classified as measures of absolute fit, measures of incremental fit, and measures of parsimonious fit (Hair, Anderson, Tatham, and Black, 1992). Measures of absolute fit determine the degree to which the proposed model predicts the observed covariance matrix. Measures of absolute fit used in this study include the chi-square statistic, the adjusted chi-square statistic, the goodness-of-fit index (GFI), and the root mean square residual (RMSR). Measures of incremental fit compare the proposed model with the null model (Hair et al., 1992). The incremental fit measure used in this study is the normed fit index (NFI). Measures of parsimonious fit attempt to determine if the

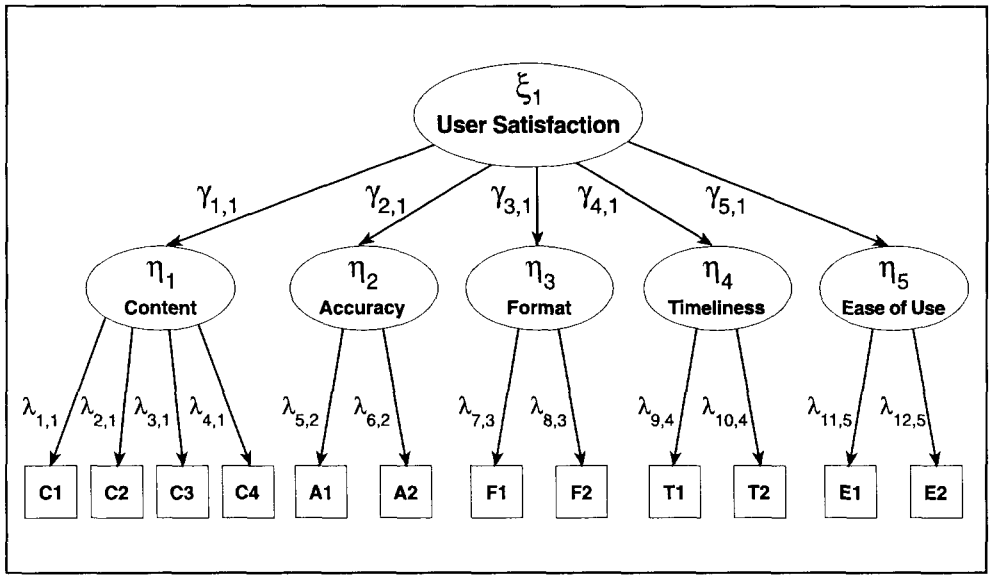

Figure 1. The user satisfaction instrument: Hypothesized model. C1: My computer system provides the precise information I need. C2: The information content from my computer system meets my needs. C3: My computer system provides reports that seem to be just about exactly what I need. C4: My computer system provides sufficient information. A1: My computer system is accurate. A2: I am satisfied with the accuracy of my computer system. F1: The output from my computer system is presented in a useful format. F2: The information from my computer system is clear. T1: I get the information I need in time from my computer system. T2: My computer system provides up-to-date information. E1: My computer system is user friendly. E2: My computer system is easy to use.

overall fit of the model has been achieved by overfitting the data. The parsimonious fit measure in this study is the adjusted goodness-of-fit index (AGFI).

A low (nonsignificant) chi-square value is desirable because such a value indicates that the actual and predicted input matrices are not statistically different. That is, the model is a good fit to the data (Bollen, 1989). The chi-square statistic for this study is 238.15 with 49 degrees of freedom ($p < .001$).

The adjusted chi-square value (χ^2/df) is used because the chi-square value is sensitive to sample size (Marsh & Hocevar, 1985). Adjusted chi-square values close to 1 indicate a well-fitting model, with values less than 5 considered acceptable (Hair et al., 1992; Bollen, 1989). The adjusted chi-square value for this study is 4.86.

The GFI is a measure of the overall degree of model fit and measures the relative amount of variance in a sample that the model predicts. The GFI is a normed index ranging from 0, indicating a poor fit, to 1, indicating a perfect fit. Acceptable values for the GFI are those greater than .9 (Hair et al., 1992; Bollen, 1989) The GFI for this study is .904.

The RMSR is a non-normed indicator of the average of the residual covariance between the observed and the estimated input matrices (Hair et al., 1992). The RMSR should be close to zero for well-fitting models. Acceptable values for the RMSR are less than .1 (Hair et al., 1992; Bollen, 1989). For this study, the RMSR was .029.

The NFI measures the fit of the proposed model to the null model (Bentler & Bonett, 1980). It represents the point at which the proposed model falls on a

scale from 0, indicating the null model, to 1, indicating a perfect fit to the data. Acceptable values of the NFI should be greater than .8 (Bentler & Bonett, 1980). The NFI for this study was .871.

The AGFI is an extension of the GFI, adjusted for degrees of freedom (Hair et al., 1992). The adjustment for degrees of freedom allows for an assessment of the parsimony of the model. If model fit has been achieved by overfitting the data with too many coefficients, the AGFI should have a low value. Values of AGFI range from 0, indicating a poor fit, to 1, indicating a perfect, yet parsimonious model. Acceptable values for the AGFI should be greater than .8 (Hair et al., 1992; Bollen, 1989). The AGFI for this study was .813.

These goodness-of-fit measures indicate that the model is a reasonable fit to our data. Even though the chi-square statistic is significant, more emphasis is placed on the adjusted χ^2, the GFI, the RMSR, the NFI, and the AGFI because chi-square values tend to be inflated by large sample sizes (Bollen, 1989). These indicators suggest that the model has a reasonable fit. Another indication of a reasonable fit is that all path coefficients in the model are significant at $p < .001$.

Reliability

Reliability refers to the extent to which the scale is free from measurement error (Nunnally, 1978). Internal consistency reliability coefficients (Cronbach alphas) were calculated to examine the reliabilities of the entire scale and each of the constructs. The Cronbach alpha for the entire 12-item scale was .93. The Cronbach alphas for the content, accuracy, ease of use, timeliness, and format constructs were .89, .65, .89, .86, and .66, respectively. These coefficients suggest that the entire scale and its underlying constructs demonstrate acceptable reliability (Nunnally, 1978).

Known Groups Analysis

The known groups criterion for construct validity signifies that scale means should discriminate across groups that theoretically would be expected to differ on the trait measured (Cronbach & Meehl, 1955). The literature provides a basis for the fact that end user computing satisfaction may vary with computer skill, computer attitudes, computer anxiety, and computer use, among others. To further validate the EUCSI and contribute to establishing a nomological network for user satisfaction, we use these four variables to develop and test seven hypotheses.

To strengthen the hypothesis testing, we not only used the entire sample in the aggregate, but we also used subsets of the sample. A recent study proposed that end users could be classified into five groups, depending on the computer tasks they perform in the course of their work (Rainer & Harrison, 1993). This study employed that scale to categorize the respondents into beginning end users, intermediate end users, advanced end users, end user facilitators, and end user infrastructure personnel. Accordingly, each hypothesis was tested using the entire sample and each of the five subgroups. Pearson correlation coefficients were calculated to test the hypotheses (Table 2). The entire sample consisted of 653 responses. This number differs from the total number of responses to the survey ($n = 776$) due to some incomplete questionnaires.

Each hypothesis employs user satisfaction as the dependent variable. Satisfaction was calculated for each respondent by summing the 5-point Likert

scores for the 12 items of the EUCSI. Consequently, user satisfaction scores could range from 12 to 60.

The relationship of computer skill to user satisfaction has not been thoroughly addressed in the literature. However, a study by Pratkanis and Greenwald (1988) suggests that satisfaction may be a function of the roles and tasks a person must perform in a given situation. Melone (1990) recommends further research on the influence of technology usage on user satisfaction.

The 32-item, computer self-efficacy scale (Murphy, Coover, & Owen, 1989) was used to measure respondents' perceptions of their capability regarding specific computer-related skills. The computer self-efficacy scale had an internal consistency reliability coefficient of .95. Respondent skill level was calculated by summing the scores of the computer self-efficacy scale items. Therefore, respondent skill scores could range from 32 to 160. The following hypothesis was proposed:

H1: There will be a significant positive correlation between user satisfaction and computer skill for the entire sample and the five subgroups.

Table 2 indicates that computer skill was significantly, positively correlated with user satisfaction for the entire sample and all five subgroups.

Igbaria and Parasuraman (1989) noted that computer attitudes can be related to user satisfaction. Mandell (1989) surveyed users and discovered that many respondents viewed computers as dehumanizing in nature.

Computer attitudes were measured by the 20-item Computer Attitude Scale (Nickell & Pinto, 1986). An EFA of this scale revealed three underlying constructs labeled as follows: negative feelings toward computers, positive feelings toward computers, and a lack of understanding of computers. The internal consistency reliability coefficients for the constructs were .82, .79, and .86, respectively. The following hypotheses were proposed:

H2: There will be a significant positive correlation between user satisfaction and positive attitudes toward computer use for the entire sample and the five subgroups.

H3: There will be a significant negative correlation between user satisfaction and negative attitudes toward computer use for the entire sample and the five subgroups.

H4: There will be a significant negative correlation between user satisfaction and a lack of understanding of computers for the entire sample and the five subgroups.

Table 2. Correlations Between User Computing Satisfaction and Related Constructs

User Computing Groups	High Anxiety	Low Anxiety	Negative Feelings	Positive Feelings	Lack of Understanding	Skill	Usage
Entire Sample (N = 653)	−.31***	.40***	−.22***	.30***	−.29***	.39***	.11**
Beginning End Users (n = 64)	−.26*	.39***	−.22*	.24*	−.21*	.50***	.36**
Intermediate End Users (n = 364)	−.32***	.37***	−.21***	.28***	−.29***	.42***	.10*
Advanced End Users (n=137)	−.31***	.42***	−.26***	.39***	−.32***	.40***	.14*
End User Facilitators (n=76)	−.25*	.31**	−.22*	.30**	−.24*	.23*	.14
End User Infrastructure Personnel (n=12)	−.64**	.51*	−.49*	.61**	−.48*	.56*	.07

*p < .05; **p < .01; ***p < .001.

Table 2 indicates that all three constructs were significantly correlated with user satisfaction in the proposed, expected directions.

Heinssen, Glass, and Knight (1987) suggested that individuals with high computer anxiety possessed lower self-confidence in their abilities and demonstrated poorer performance outcomes than subjects with low computer anxiety. Igbaria and Parasuraman (1989) noted that computer anxiety can be related to dissatisfaction with the computer system.

Computer anxiety was measured by the 19-item Computer Anxiety Rating Scale (Heinssen, Glass, & Knight, 1987). An EFA of this scale demonstrated two underlying constructs labeled high anxiety toward computer use and low anxiety toward computer use. The internal consistency reliability coefficients for the two constructs were .84 and .85, respectively. The following hypotheses were proposed:

H5: There will be a significant positive correlation between user satisfaction and low anxiety toward computer use for the entire sample and the five subgroups.

H6: There will be a significant negative correlation between user satisfaction and high anxiety toward computer use for the entire sample and the five subgroups.

Table 2 shows that both constructs were significantly correlated to user satisfaction in the proposed, expected directions.

Baroudi, Olson, and Ives (1986) and Igbaria and Nachman (1990) found strong relationships between user satisfaction and computer usage. Computer usage of a variety of computer hardware (supercomputers, mainframes, minicomputers, workstations, microcomputers) was measured by 5-point Likert scales ranging from 1, "I do not use this computer at all", to 5, "I use this computer frequently every day". Each respondent's usage score was the sum of the usage scores across the five types of computer hardware. Therefore, each respondent's usage score could range from 5 to 25.

The following hypothesis was proposed:

H7: There will be a significant positive correlation between user satisfaction and computer use for the entire sample and the five subgroups.

Table 2 shows that usage is significantly, positively correlated with user satisfaction for the entire sample and three of the subgroups. End user facilitators and end user infrastructure personnel demonstrated nonsignificant correlations.

DISCUSSION

The values of the goodness-of-fit indices obtained from the confirmatory factor analysis indicate that the second-order model of the EUCSI provides a reasonable fit to the data. These findings suggest the presence of a higher-order factor, user satisfaction, that explains a portion of the covariance among the first order factors (content, accuracy, format, timeliness, and ease of use). The reasonable fit of the model to the data supports the construct validity of the instrument developed by Doll and Torkzadeh (1991) when used as a general measure of user satisfaction. The Cronbach alpha coefficients indicate that the

five constructs and the entire instrument demonstrate adequate reliability as used in this study.

The known groups analysis provides additional evidence supporting the construct validity of the instrument as used in this study. The purpose of this article was to show that the EUCSI could be used across all organizational personnel who used computers in their work and across all applications and computer systems they employed. The correlations supported all hypotheses across all subgroups, except for computer usage in the end user facilitators and end user infrastructure personnel subgroups. The correlations furnish very strong support for the construct validity of the instrument as used in this study.

Implications for Theory and Research

The use of the EUCSI as a general measure of user satisfaction has a variety of implications for theory and research. This study defined and delineated the construct, general user computing satisfaction. Doll and Torkzadeh (1991) suggested that general satisfaction may be confounded by a variety of closely related constructs. The known groups analysis in this study demonstrated that general satisfaction may be closely related to several computing-related variables (e.g., computer attitudes, computer anxiety, etc). These relationships should be studied in future research to determine the direction of causality.

Igbaria and Parasuraman (1989) provided researchers with a path model of the relationship between computer attitudes and computer anxiety. Their model could be more fully developed to better understand the complex relationships between these constructs and general user computing satisfaction.

An additional direction for future research on the topic of general user computing satisfaction would be to examine the stability of the construct. Some researchers (e.g., Pancer, George, & Gebotys, 1992) suggest that user satisfaction is an attitude that may be affected by computer training (i.e., user computing satisfaction is a state). However, other research suggests that such attitudes may be very stable and unchanging, thereby representing more of an individual trait than state.

For example, Staw and Ross (1985) found that job attitudes were consistent traits within individuals across time and different jobs. User computing satisfaction should be examined in a similar fashion to determine if in fact the construct is an attitude that may be altered by training, education, or changes in hardware or software, or if the construct is a relatively stable disposition or trait within the individual user that does not change across organizations, systems, or training. Research examining the state versus trait argument could have considerable implications for training and development of end users.

Implications for Practicing Managers

The use of the EUCSI as a general measure of user computing satisfaction has implications for management of end user computing in organizations. The management of end user computing is often examined at multiple levels: (a) the individual level, including the job-related computing activities of the individual employee, and (b) the organizational level encompassing education, training, standards, security, controls, etc. for the organization as a whole (Rainer & Harrison, 1993).

The use of the EUCSI as a general measure of user computing satisfaction can be particularly beneficial to information systems managers at the organizational level. This measure will allow managers to assess computer systems across departments, users, and applications to gain an overall view of user satisfaction.

This use of the EUCSI as a general measure does not contradict the original use of the instrument by Doll and Torkzadeh (1991), which measured application-specific computing satisfaction. Using the scale as a general measure as well as an application-specific measure could help the information systems manager gain a broader perspective of user satisfaction with systems and applications across the organization.

CONCLUSIONS

We make no attempt to generalize beyond this particular university environment. The instrument should be used in a variety of other settings (e.g., public and private sector enterprises) to assess its external validity. In addition, common method variation is a limitation of this study.

Practicing managers can use the EUCSI as a quick, easy-to-use, easy-to-understand indication of user satisfaction throughout an organization. This measure of user satisfaction can give these managers an idea as to the effectiveness of the computing resource in the organization. While the contextual richness of application-specific satisfaction may need to be measured using the EUCSI in its original fashion, the use of the instrument as a general measure will provide generalizability for an overall view.

This research suggests an additional manner in which the EUCSI developed by Doll and Torkzadeh (1991) may be utilized by practicing managers and researchers. Instead of using the EUCSI exclusively to measure user satisfaction with a specific application, the instrument may also be used as an easy-to-complete measure of general user satisfaction across all organizational personnel who use computers in their work and across all applications they employ. The current form of the EUCSI may not be final. For example, equity issues are not included (Joshi, 1990). However, the instrument deserves additional validation studies.

NOTE

[1]The authors contributed equally to the design and execution of this research project.

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